06/12/2023

Code-A_Phase-1

Aakash + BYJU'S

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MM : 720

NCERT Booster Test Series_CF+OYM_2023-24_Test-01A

Time : 200 Min.

Topics Covered:
Physics : Thermodynamics, Kinetic Theory, Oscillations, Waves, Electromagnetic Induction
Chemistry : Organic Chemistry: Some Basic Principles and Techniques, Hydrocarbons
Botany : Photosynthesis in Higher Plants, Respiration in Plants, Plant Growth and Development
Zoology : Animal Kingdom, Structural Organisation in Animals: Animal Morphology of Cockroach, Earthworm, Frog, Human Health and Disease

General Instructions :

1. There are two sections in each subject, i.e. **Section-A & Section-B**. You have to attempt all **35 questions** from Section-A & only **10 questions** from Section-B out of **15.**
2. Each question carries **+4 marks.** For every wrong response, **–1 mark** shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
3. Use blue/black ballpoint pen only to darken the appropriate circle.
4. Mark should be dark and completely fill the circle.
5. Dark only one circle for each entry.
6. Dark the circle in the space provided only.
7. Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

SECTION-A

1. The time period of oscillations of seconds pendulum mounted in a cabin that is freely falling under gravity, is

(1) 2 s
(2) 1 s
(3) Infinite
(4) Zero

2. The time period of oscillation for the periodic motion given by

$Y = \sin\omega t + \sin 2\omega t + \cos\omega t$

is

(1) $\frac{2\pi}{\omega}$
(2) $\frac{\pi}{\omega}$
(3) $\frac{\pi}{2\omega}$
(4) $\frac{4\pi}{\omega}$

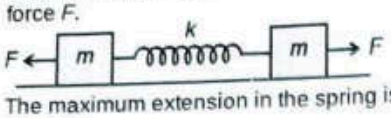
3. Motion of a kink in a longitudinal spring produced by displacing one end of the spring sideways is

(1) Longitudinal only
(2) Both longitudinal and transverse
(3) Transverse only
(4) Neither transverse nor longitudinal

4. A steel wire of length 0.75 m has a mass of 5.0×10^{-3} kg. If the wire is under a tension of 60 N, then the speed of transverse waves on the wire is

(1) $30\sqrt{10}$ m s^{-1}
(2) $300\sqrt{10}$ m s^{-1}
(3) $0.3\sqrt{10}$ m s^{-1}
(4) $3\sqrt{10}$ m s^{-1}

5. The figure given shows a spring of force constant k attached to mass m at either end. Each end of spring is stretched by the same force F.



The maximum extension in the spring is

(1) $\frac{F}{k}$

(2) $\frac{F}{2k}$

(3) $\frac{2F}{k}$

(4) $\frac{4F}{k}$

6. The speed of sound in air is given by $v = \sqrt{\frac{\gamma P}{\rho}}$

Where γ is equal to

(1) $\frac{3}{5}$

(2) $\frac{7}{5}$

(3) 3

(4) 4

7. A pipe 30.0 cm long, is open at both ends, which harmonic of pipe resonates at 2.2 kHz source? Speed of sound in air is 330 ms^{-1}.

(1) 1

(2) 2

(3) 3

(4) 4

8. The molar specific heat capacity of solids is

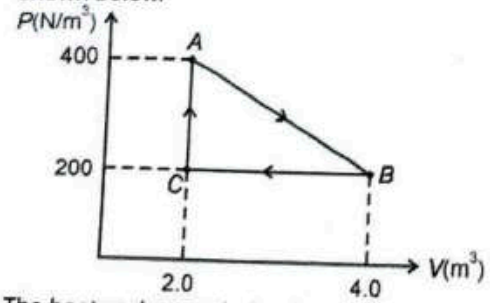
(1) $3R$

(2) $\frac{3}{2}R$

(3) $\frac{5}{2}R$

(4) $4R$

9. P-V diagram of a one mole monatomic gas undergoing cyclic process $ABCA$ is as shown below.



The heat exchange during the cyclic process $ABCA$ is

(1) 200 J

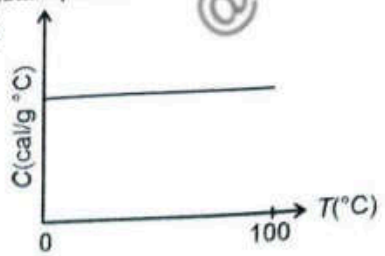
(2) 300 J

(3) 400 J

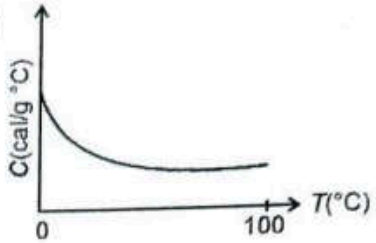
(4) 500 J

10. Choose the correct graph of specific heat capacity (C) of water with temperature (T) at constant pressure.

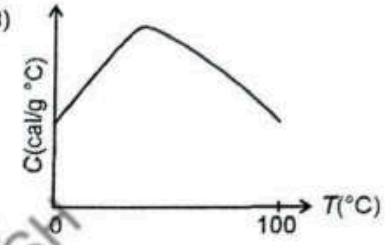
(1)



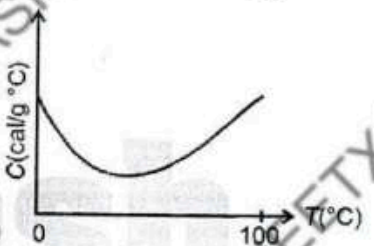
(2)



(3)



(4)



11. In a cyclic process, the work done on the system over complete cycle

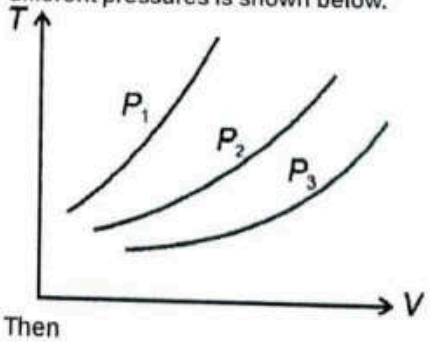
(1) Depends upon the path

(2) Is always positive

(3) Is always negative

(4) Is always zero

12. Experimental T-V diagram of CO_2 gas at different pressures is shown below.



Then

(1) $P_1 = P_2 = P_3$

(2) $P_1 > P_2 > P_3$

(3) $P_1 < P_2 < P_3$

(4) None of the above

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13. A flask contains argon and chlorine in the ratio of 2 : 1 by mass. The temperature of mixture is 127°C. If the atomic mass of Argon is 40 u and molecular mass of chlorine gas is 70 u then the ratio of root mean square (V_{rms}) of the molecules of two gases is
(1) 1.77
(2) 1.87
(3) 1.32
(4) 1.67

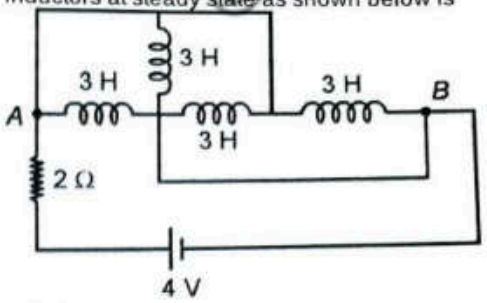
14. *n* moles of diatomic gas in a cylinder are at a temperature *T*. Heat is supplied to the cylinder such that temperature of the gas becomes 2*T* and diatomic gas get converted into monatomic gas. The change in total kinetic energy of the gas is
(1) $\frac{7}{2}nRT$
(2) $\frac{5}{2}nRT$
(3) $\frac{3}{2}nRT$
(4) nRT

15. When the temperature of an ideal monatomic gas is increased
(1) Its molecular kinetic energy increases while its potential energy decreases
(2) Its molecular kinetic energy and potential energy both increases
(3) It molecular potential energy decreases
(4) It molecular kinetic energy increases

16. 2 moles of monatomic and 1 mole of diatomic gas are mixed together at constant temperature. The value of γ for the mixture is
(1) $\frac{17}{11}$
(2) $\frac{17}{13}$
(3) $\frac{19}{11}$
(4) $\frac{19}{13}$

17. A very small square loop of side *a* is initially coplanar and concentric with much larger fixed circular loop of radius *b*. A constant current *I* flows through the bigger loop. The smaller loop is rotated with a constant angular speed ω about the diameter of bigger loop. The emf induced in the smaller loop as a function of time *t* is
(1) $\frac{a^2\mu_0 I}{2b}\omega\cos\omega t$
(2) $\frac{a^2\mu_0 I}{2b}\omega\sin(\omega^2 t^2)$
(3) $\frac{a^2\mu_0 I}{2b}\omega\sin(\omega t)$
(4) $\frac{a^2 I\mu_0}{2b}\omega\sin^2(\omega t)$

18. The energy stored in the combination of inductors at steady state as shown below is

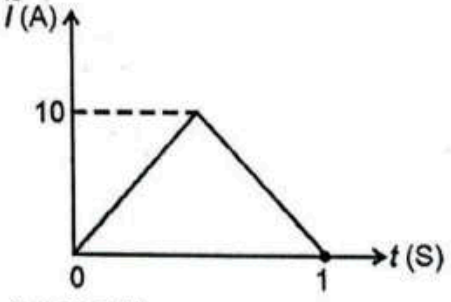


(1) 3 J
(2) $\frac{3}{2}$ J
(3) 2 J
(4) $\frac{5}{2}$ J

19. The magnetic flux through a loop of resistance *R* changes as $\Delta\varphi$ in time Δt then SI unit of $\frac{\Delta\phi}{R}$ is (symbol have their usual meaning)
(1) A s
(2) A s^{-1}
(3) J V^{-1}
(4) Both (1) and (3)

20. Whenever there is relative motion between a coil and a magnet the magnitude of induced emf set up in coil depends upon
(1) Relative speed between the coil and magnet
(2) Magnetic moment of coil
(3) Number of turns in the coil
(4) All of the above

21. In a coil of resistance 100 Ω, a current is induced by changing the magnetic flux through it as shown in the figure. The magnitude of change in flux through the coil is



(1) 250 Wb
(2) 200 Wb
(3) 500 Wb
(4) 400 Wb

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22. The displacement of a particle in SHM is given by

$x = 10\sin\left(2t - \frac{\pi}{6}\right)m$

When the displacement is 8 m, the velocity of the particle is

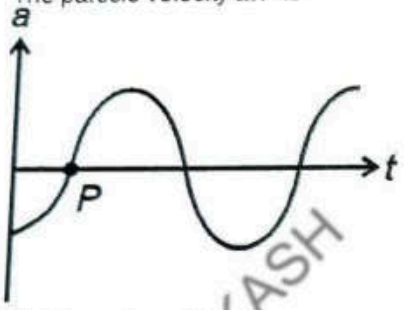
(1) $8\ m\ s^{-1}$

(2) $12\ m\ s^{-1}$

(3) $20\ m\ s^{-1}$

(4) $10\ m\ s^{-1}$

23. The figure shows the plot of acceleration (a) with time (t) for a particle executing simple harmonic motion about origin along x axis. The particle velocity at P is



(1) Towards positive x

(2) Towards negative x

(3) Zero

(4) May be towards positive x or towards negative x

24. The equation $y = A\sin^2\left(4\pi nt - \frac{2\pi x}{\lambda}\right)$ represent a wave of

(1) Amplitude A, frequency n and wavelength λ

(2) Amplitude $\frac{A}{2}$, frequency $4n$ and wavelength $\frac{\lambda}{2}$

(3) Amplitude $\frac{A}{2}$, frequency $2n$ and wavelength λ

(4) Amplitude A, frequency $2n$ and wavelength λ

25. When the wavelength of sound wave changes from 1 m to 1.01 m, the number of beats heard are 3, the velocity of the sound is

(1) $300\ m\ s^{-1}$

(2) $330\ m\ s^{-1}$

(3) $390\ m\ s^{-1}$

(4) $303\ m\ s^{-1}$

26. If the slope of isothermal process is x then the slope of adiabatic process for diatomic gas is

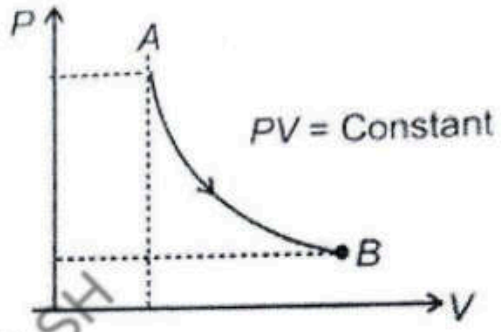
(1) x

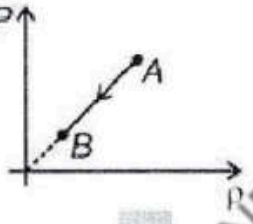
(2) $\frac{4}{3}x$

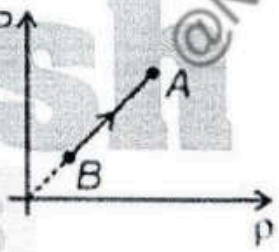
(3) $\frac{5}{3}x$

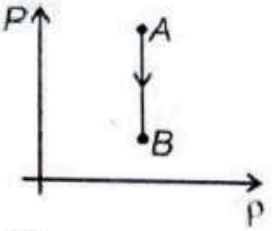
(4) $\frac{7}{5}x$

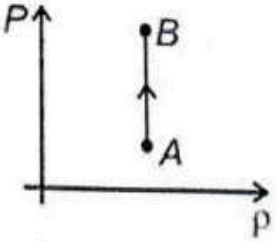
27. Consider P-V diagram for fixed mass of an ideal gas shown in figure. Choose correct corresponding pressure-density (P-ρ) diagram.



(1)


(2)


(3)


(4)


28. Mean free path of molecules of a gas is $\frac{1}{\sqrt{2}\pi}$ cm. If number density of gas is $10^8/cm^3$, then the diameter of the gas molecule is

(1) 10^{-4} cm

(2) 10^4 cm

(3) 10^{-8} cm

(4) 10^{-6} cm

29. Temperature of a gas is increased from 127°C to 527°C. The rms speed of gas molecules will increase to a factor of

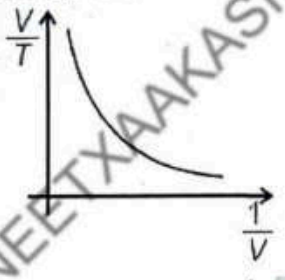
(1) 2

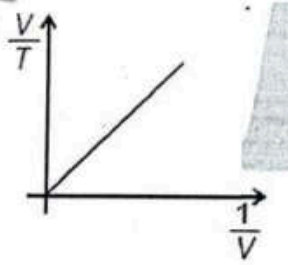
(2) 4

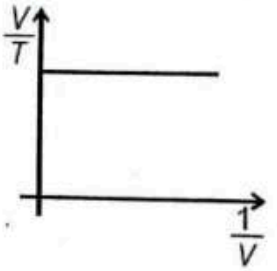
(3) $\sqrt{2}$

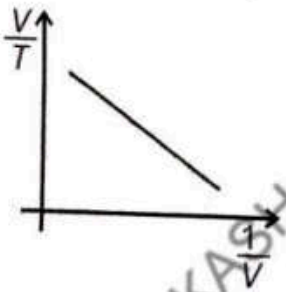
(4) $\sqrt{3}$

30. The correct graph between $\frac{V}{T}$ and $\frac{1}{V}$ for an ideal gas at constant pressure is

(1) 

(2) 

(3) 

(4) 

31. The speed of six molecules of a gas are 2, 3, 7, 8, 9, 10 units. The average speed for these molecules is

(1) 6 units

(2) 2.9 units

(3) 6.5 units

(4) 7.5 units

32. An ideal spring with spring constant K is suspended from the ceiling and a block of mass m is connected to its lower end. The mass is released with the spring initially unstretched. The maximum velocity of the block is

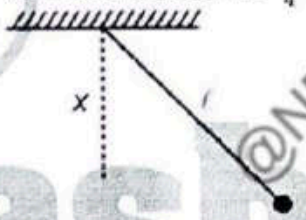
(1) $g\sqrt{\frac{K}{m}}$

(2) $g\sqrt{\frac{m}{K}}$

(3) $\frac{g}{\sqrt{mK}}$

(4) $g\sqrt{2mK}$

33. Time period of small oscillation of simple pendulum of length ℓ is T. Now, if a small obstacle is placed at a distance x below the point of suspension, the time period for new oscillations, becomes $\frac{3T}{4}$. Find x.



(1) $\frac{\ell}{4}$

(2) $\frac{\ell}{2}$

(3) $\frac{3\ell}{4}$

(4) $\frac{3\ell}{8}$

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34. A simple pendulum with a metallic bob has a time period 10 s. The bob is now immersed in a non-viscous liquid of density $\frac{1}{3}$ that of metallic bob. The time period of the same pendulum becomes

(1) 10 s

(2) $\frac{10}{\sqrt{2}}$ s

(3) $10\sqrt{\frac{3}{2}}$ s

(4) $10\sqrt{3}$ s

35. A particle is executing SHM along a straight line starts from mean position, its velocity at position of 3 cm and 4 cm is 4 cm/s and 3 cm/s respectively then its time period (in second) is

(1) $\frac{\pi}{2}$

(2) π

(3) $4\pi^2$

(4) 2π

SECTION-B

36. If a current of 5 A reduces to zero in one second in a coil, and the emf induced in the coil is 10 V, then the self-inductance of the coil is

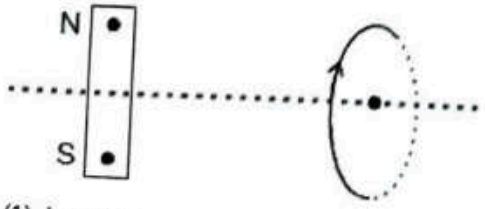
(1) 1 H

(2) 4 H

(3) 3 H

(4) 2 H

37. As shown in figure, a permanent magnet and current carrying coil are placed. If the coil is moved toward magnet, then current in coil



(1) Increase

(2) Decrease

(3) Remains same

(4) First increase then decrease

38. An ideal gas at 27°C is compressed adiabatically to $\frac{8}{27}$ of its original volume. If $\gamma = \frac{5}{3}$, then rise in temperature is

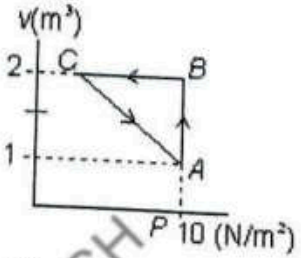
(1) 375 K

(2) 475 K

(3) 575 K

(4) 675 K

39. An ideal gas is taken through the cycle $A \rightarrow B \rightarrow C \rightarrow A$ as shown in the figure. If the net heat supplied to the gas in the cycle is 3 J, the work done by the gas in the process $C \rightarrow A$ is



(1) −7 J

(2) −10 J

(3) −15 J

(4) −20 J

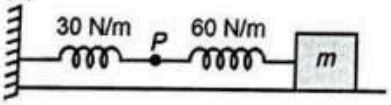
40. A diatomic gas does a work of $\frac{Q}{4}$ when heat Q is supplied to gas. The molar heat capacity of gas is
(1) $\frac{6}{5}R$
(2) $\frac{10}{3}R$
(3) $\frac{5}{2}R$
(4) $\frac{5}{3}R$

41. Calculate average kinetic energy of oxygen gas molecules at 0°C. (Given $K_B = 1.38 \times 10^{-23}$ J/K)
(1) 9.42×10^{-21} J
(2) 10×10^{21} J
(3) 4.6×10^{-9} J
(4) 5.10×10^{-41} J

42. A : Average velocity of gas molecules is zero.
R : Average speed of gas molecules is non zero at room temperature.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

43. Two simple harmonic motions are represented by the equations $y_1 = 5\sin\left(4\pi t + \frac{\pi}{6}\right)$ and $y_2 = 3\sin(2\pi t) + 4\cos(2\pi t)$. Then the ratio of their amplitude is
(1) 1 : 1
(2) 2 : 1
(3) 5 : 7
(4) 5 : 1

44. Two spring of spring constant 30 N/m and 60 N/m are joined in series and a mass m is attached as shown in the figure. If the amplitude of oscillation of the mass is 3 cm, then what is the amplitude of oscillation of P?



(1) 1 cm
(2) 2 cm
(3) 3 cm
(4) 1.5 cm

45. **Statement (A):** Generally oscillatory motion is periodic motion.
Statement (B): Every periodic motion need not be on oscillatory motion.
(1) Only statement (A) is correct
(2) Only statement (B) is correct
(3) Both statement (A) and (B) are correct
(4) Neither statement (A) nor statement (B) are correct

46. Two sources produce sound waves of equal amplitudes and travelling along the same direction producing 18 beats in 3 seconds. If one source has a frequency of 341 Hz, the frequency of the other source may be
(1) 329 or 353 Hz
(2) 335 or 347 Hz
(3) 338 or 344 Hz
(4) 332 or 350 Hz

47. A wave of frequency 600 Hz has velocity 300 m/s. The distance between two nearest point 60° out phase is, nearly
(1) 6 cm
(2) 12 cm
(3) 6.66 cm
(4) 8.33 cm

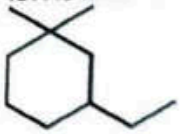
48. A closed organ pipe has frequency f. If its length is doubled and radius is halved, then its fundamental frequency will become
(1) Halved
(2) Doubled
(3) Tripled
(4) Quadrupled

49. The coefficient of coupling between two coils having self-inductance 16 H and 9 H and coefficient of mutual inductance 9 H, is
(1) 0.65
(2) 0.75
(3) 0.8
(4) 1.0

50. The speed of longitudinal wave in a metal rod having Young's modulus 1×10^{10} N/m^2 and density as 2.5×10^3 kg/m^3 is
(1) 1000 m/s
(2) 2000 m/s
(3) 3000 m/s
(4) 4000 m/s

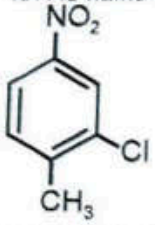
SECTION-A

51. IUPAC name of the compound



(1) 3-Ethyl-1, 1-dimethylcyclohexane
(2) 1-Ethyl-3, 3-dimethylcyclohexane
(3) 5-Ethyl-1, 1-dimethylcyclohexane
(4) 1-Ethyl-5, 5-dimethylcyclohexane

52. IUPAC name of the compound



(1) 4-Methyl, 5-chloronitrobenzene
(2) 4-Methyl, 3-chloronitrobenzene
(3) 1-Chloro-2-methyl-5-nitrobenzene
(4) 2-Chloro-1-methyl-4-nitrobenzene

53. Propanone and propanal are _____ isomer of each other.
(1) Chain isomer
(2) Position
(3) Functional group
(4) Metamer

54. Which of the following reagent is not a nucleophile?
(1) R_3N
(2) $H_2\ddot{O}$
(3) $\ominus_{CN}$
(4) $\oplus_{CH_3}$

55. Which of the following stability order for carbocation is correct?
(1) $(CH_3)_3C^{\oplus} > (CH_3)_2\overset{\oplus}{C}H > \overset{\oplus}{C}H_3 > CH_3\overset{\oplus}{C}H_2$
(2) $(CH_3)_2\overset{\oplus}{C}H > (CH_3)_3C^{\oplus} > CH_3\overset{\oplus}{C}H_2 > \overset{\oplus}{C}H_3$
(3) $\overset{\oplus}{C}H_3 > CH_3\overset{\oplus}{C}H_2 > (CH_3)_2\overset{\oplus}{C}H > (CH_3)_3C^{\oplus}$
(4) $(CH_3)_3C^{\oplus} > (CH_3)_2\overset{\oplus}{C}H > CH_3\overset{\oplus}{C}H_2 > \overset{\oplus}{C}H_3$

56. Which among the following behaves as electron donating group when attached with benzene?
(1) NO_2
(2) –CN
(3) –COOH
(4) $-CH_3$

57. For the given reactions (a) and (b), major products [A] and [B] respectively are

(a) $CH_4 + O_2 \xrightarrow[100\ atm]{Cu\ |523\ K} [A]$

(b) $CH_4 + O_2 \xrightarrow[\Delta]{M_{O_2}O_3} [B]$

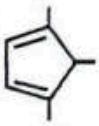
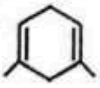
(1) Methanol and methanal
(2) Methanal and oxalic acid
(3) Glyoxal and methanal
(4) Glyoxal and methanol

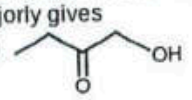
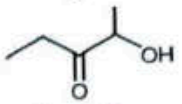
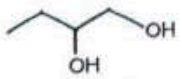
58. Considering the following statements
(a) PbS is yellow in colour
(b) On treating sodium fusion extract with sodium nitroprusside, appearance of a violet colour indicates the presence of sulphur in the compound
(c) AgBr is sparingly soluble in ammonium hydroxide.
The correct statements are
(1) (b) and (c) only
(2) (a) and (b) only
(3) (a), (b) and (c)
(4) (a) and (c) only

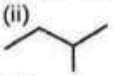
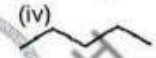
59. In sulphur estimation 2.5 g of an organic compound gave 2.097 g of barium sulphate. The percentage of sulphur in the compound is (Molar mass of $BaSO_4$ = 233 g mol^{-1})
(1) 32.4%
(2) 24.5%
(3) 11.5%
(4) 18.2%

60. Meta directing group towards electrophilic substitution reaction of benzene is
(1) $-NHCOCH_3$
(2) $-C_2H_5$
(3) $-CN$
(4) $-OCH_3$

61. Which among the following is not an aromatic species?
(1) 
(2) 
(3) 
(4) 

62. Which compound on reductive ozonolysis will give glyoxal as one of the products?
(1) 
(2) 
(3) 
(4) 

63. But-1-ene on reaction with Baeyer's reagent, majorly gives
(1) 
(2) 
(3) 
(4) 

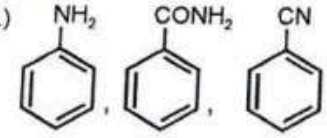
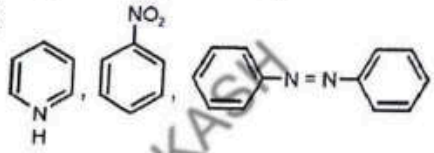
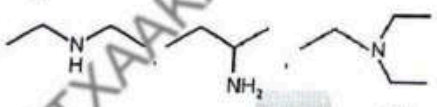
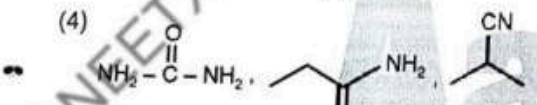
64. Correct order of boiling point of the given compounds is
(i) 
(ii) 
(iii) 
(iv) 
(1) (iv) > (i) > (ii) > (iii)
(2) (i) > (iv) > (ii) > (iii)
(3) (iii) > (i) > (iv) > (ii)
(4) (iii) > (ii) > (i) > (iv)

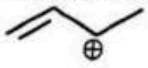
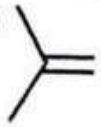
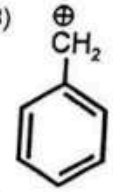
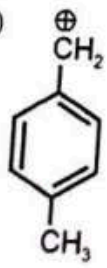
During estimation of nitrogen present in an organic compound by Kjeldahl's method, the ammonia evolved from 2.0 g of organic compound was neutralized by 25 mL of 1 M H_2SO_4 solution. The percentage of nitrogen in the compound is
(1) 35%
(2) 22%
(3) 14%
(4) 42%

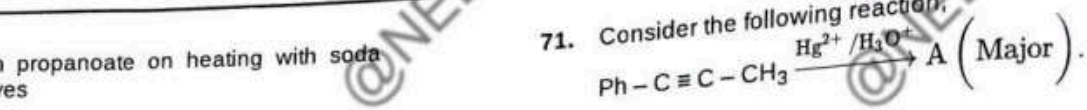
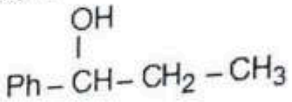
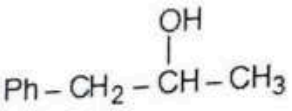
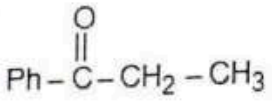
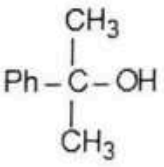
66. Given below are the two statements
Statement I: Alkyl fluorides on reduction with zinc and dilute HCl gives alkanes.
Statement II: Wurtz reaction is done in presence of dry ether.
In light of the above statements, choose the **correct** answer.
(1) Statement I is correct but statement II is incorrect.
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are correct
(4) Both statement I and statement II are incorrect

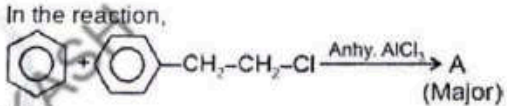
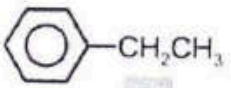
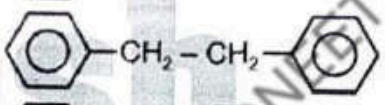
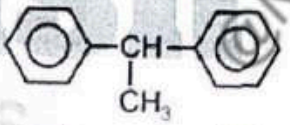
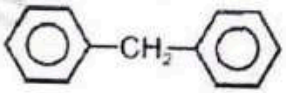
67. Sodium propanoate on heating with soda lime gives
(1) Ethane
(2) Ethene
(3) Propene
(4) Propane

68. Colour of the complexes $[Fe(SCN)]^{2+}$ and $[Fe(CN)_5NOS]^{4-}$ respectively are
(1) Blood red and Prussian blue
(2) Orange and Green
(3) Blood red and Violet
(4) Yellow and Blood red

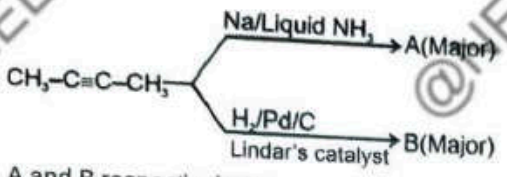
69. Which among the given sets of compounds, nitrogen can not be estimated by Kjeldahl method?
(1) 
(2) 
(3) 
(4) 

70. The species which does not show hyperconjugation effect is
(1) 
(2) 
(3) 
(4) 

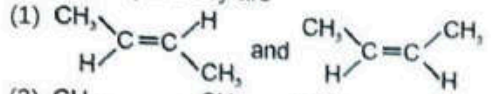
71. Consider the following reaction:

Identify A.
(1) 
(2) 
(3) 
(4) 

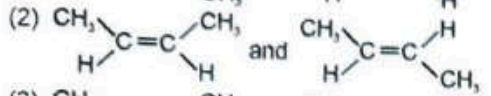
72. In the reaction,

A is
(1) 
(2) 
(3) 
(4) 

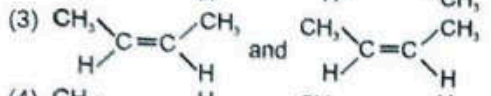
73. The compound which is least reactive towards Friedel-Crafts reaction is
(1) Ethylbenzene
(2) Chlorobenzene
(3) Nitrobenzene
(4) Benzene

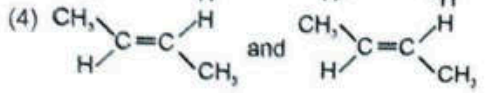
74. 

A and B respectively are

(1) 

(2) 

(3) 

(4) 

75. Most acidic compound among the following is

(1) $CH_3 - C \equiv C - CH_3$

(2) $CH_3 - C \equiv CH$

(3) $HC \equiv CH$

(4) $H_2C = CH_2$

76. Which among the following elements does not give Lassaigne's test?

(1) Phosphorus

(2) Chlorine

(3) Boron

(4) Sulphur

77. Assertion : Inductive effect is a permanent effect.

Reason : Inductive effect weakens steadily with increasing distance from the substituent.

In light of the above statements, choose the correct answer

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

78. In the detection of P, yellow precipitate obtained is due to formation of

(1) $(NH_4)_3PO_4$

(2) $(NH_4)_3PO_4.MoO_3$

(3) $(NH_4)_3PO_4.12MoO_3$

(4) $12(NH_4)_3PO_4.MoO_3$

79. The most stable species, among the following is

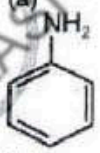
(1) 

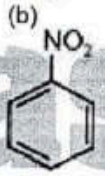
(2) 

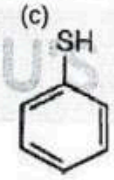
(3) 

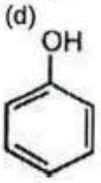
(4) 

80. Lassaigne's extract of compound 'X' was taken in a test tube and was made alkaline by adding NaOH solution. On further adding freshly prepared ferrous sulphate followed by boiling (the reaction mixture), we get the formation of Prussian blue coloured solution. Compound 'X' can be

(a) 

(b) 

(c) 

(d) 

(1) Only (a)

(2) Only (d)

(3) Both (a) & (c)

(4) Both (a) & (b)

81. Total number of resonating structure of nitrobenzene are

(1) 3

(2) 4

(3) 6

(4) 5

82. Which among the following is the strongest acid?

(1) CH_3COOH

(2) FCH_2COOH

(3) $F_2CHCOOH$

(4) F_3CCOOH

83. The enol form of hept-5-en-2-one contains

(1) 17σ-bonds and 2π-bonds

(2) 19σ-bonds and 2π-bonds

(3) 15σ-bonds and 1π-bond

(4) 21σ-bonds and 2π-bonds

84. In Dumas method for estimation of nitrogen, 0.25 g of an organic compound gave 40 mL of nitrogen collected at 300 K temperature and 725 mm pressure. If the aqueous tension at 300 K is 25 mm, the percentage of nitrogen in the compound is

(1) 15.76

(2) 17.36

(3) 18.20

(4) 16.76

85. 0.5 g of an organic compound on combustion gives 0.2 g of CO_2 and some water. The percentage of carbon in the compound is

(1) 5.1%

(2) 10.9%

(3) 23.7%

(4) 43%

SECTION-B

86. Lassaigne's test for nitrogen is negative for which of the following compounds?

(1) Hydrazine

(2) Acetamide

(3) Urea

(4) Ethylamine

87. Which of the following alkanes cannot be prepared by Kolbe's electrolytic method?

(1) CH_4

(2) C_2H_6

(3) C_4H_{10}

(4) C_6H_{14}

88. The alkene which gives only one product on reaction with HBr is

(1) 2, 3-dimethylbut-2-ene

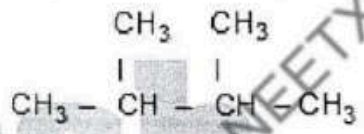
(2) Pent-2-ene

(3) 2-methylbut-2-ene

(4) Hex-2-ene

89. n-Hexane on heating to 773 K at 10–20 atm pressure in presence of vanadium pentaoxide forms

(1) $CH_2 = CH - CH_2 - CH_2 - CH_2 - CH_3$

(2) 

(3) 

(4) 

90. The alkane which on monochlorination gives maximum number of products (structural isomers) is

(1) 2-2-dimethylpropane

(2) 2-methylpropane

(3) 2-methylbutane

(4) Pentane

91. The technique used to separate chloroform and aniline is

(1) Distillation

(2) Crystallisation

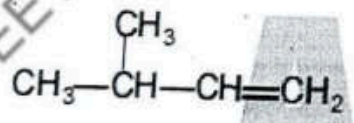
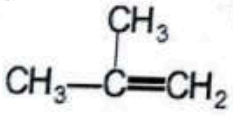
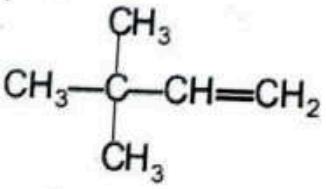
(3) Chromatography

(4) Steam distillation

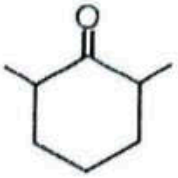
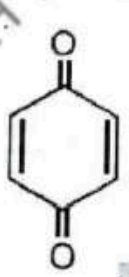
92. Which among the following is a temporary effect?
(1) Inductive effect
(2) Electromeric effect
(3) Resonance
(4) Hyperconjugation

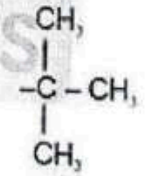
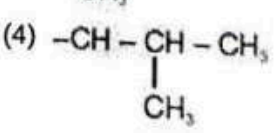
93. Gas produced at cathode in Kolbe's electrolysis of CH_3COONa (aq) is
(1) CH_4
(2) H_2
(3) CO_2
(4) C_2H_6

94. The hybridisation state of negatively charged carbon atom in $CH_3 - \overset{\ominus}{C}H_2$ is
(1) *sp*
(2) sp^2
(3) sp^3
(4) sp^3d

95. Compound having maximum number of hyperconjugative structures is
(1) $CH_3CH=CH_2$
(2) 
(3) 
(4) 

96. Total number of structural isomers of C_6H_{14} is
(1) 4
(2) 5
(3) 6
(4) 3

97. The compound which will not show keto-enol tautomerism is
(1) 
(2) 
(3) 
(4) 

98. The strongest +I effect is shown by
(1) $-CH_2CH_2CH_2CH_3$
(2) $-CH_2CH_3$
(3) 
(4) 

99. Which of the following is most stable carbanion?

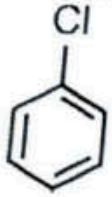
(1) $CH \equiv C^{\ominus}$

(2) $(CH_3)_2CH^{\ominus}$

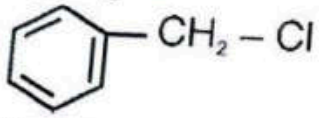
(3) $CH_2 = CH^{\ominus}$

(4) $(CH_3)_3C^{-}$

100. The C – Cl bond length is shortest in

(1) 

(2) $CH_3 - CH_2 - Cl$

(3) 

(4) $CH_2 = CH - CH_2 - Cl$

BOTANY

SECTION-A

101. Read the following statements, state them as true **(T)** or false **(F)** and select the **correct** option.

(a) There is a linear relationship between incident light and CO_2 fixation rates at high light intensities.

(b) CO_2 is the major limiting factor for photosynthesis.

(c) Dark reaction in photosynthesis is more temperature sensitive than light reaction.

(d) The effect of water on photosynthesis as a factor is more through its effect on the plant, rather than directly.

	(a)	(b)	(c)	(d)
(1)	F	F	F	T
(2)	F	F	T	T
(3)	F	T	T	T
(4)	T	F	F	F

(1) (1)

(2) (2)

(3) (3)

(4) (4)

102. **Statement A** : C_4 plants respond to high light intensities, lack photorespiration and have greater productivity of biomass.

Statement B : C_4 plants show saturation at about 360 μIL^{-1} of CO_2 concentration while C_3 plants saturate only below 450 μIL^{-1}.

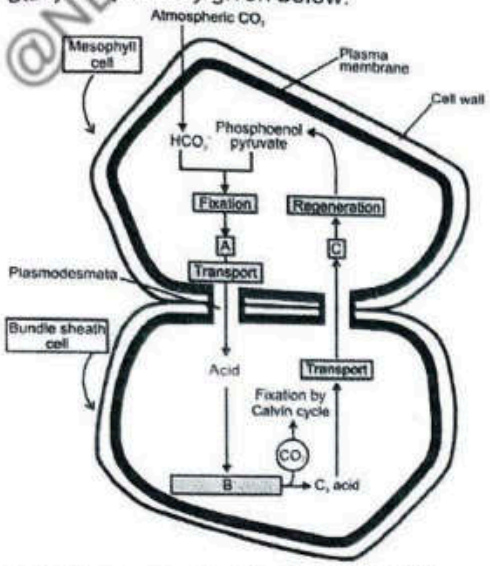
(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Statement A is correct but statement B is incorrect

(4) Statement B is correct but statement A is incorrect

103. Study the pathway given below:



Select the **correct** option for A, B and C.

(1) **A–** C_3 acid, **B–** C_4 acid, **C–**Fixation

(2) **A–** C_4 acid, **B–** Fixation, **C–** C_3 acid

(3) **A–** C_4 acid, **B–** Decarboxylation, **C–** C_3 acid

(4) **A–** C_3 acid, **B–** C_4 acid, **C–** Decarboxylation

104. Select the **odd** one out w.r.t. role of water in plants.

(1) It helps in plant growth and development

(2) It acts as a source of energy

(3) It provides the medium for enzymatic activities needed for growth

(4) It helps plant cells to grow in size by cell enlargement

105. Match the following columns and select the **correct** option.

	Column-I		Column-II
a.	Phosphoglycolate	(i)	Primary CO_2 acceptor in C_3 plants
b.	Phosphoenol pyruvate	(ii)	2 carbon compound
c.	RuBisCO	(iii)	Primary CO_2 acceptor in C_4 plants
d.	RuBP	(iv)	Most abundant enzyme on earth

(1) a(iii), b(iv), c(i), d(ii)

(2) a(ii), b(iii), c(iv), d(i)

(3) a(ii), b(iv), c(i), d(iii)

(4) a(iii), b(ii), c(iv), d(i)

106. Read the following statements and select the option for the **correct** statement(s).
(A) Fatty acids would be broken down to acetyl CoA before entering the respiratory pathway when it is used as a substrate.
(B) Pure proteins or fats are never used as respiratory substrates.
(C) Respiratory pathway is involved in both anabolism and catabolism.

(1) Only (A) and (B)

(2) Only (A) and (C)

(3) Only (C)

(4) All (A), (B) and (C)

107. ATP synthesis in chloroplast takes place

(1) In the intermembrane space

(2) When there is movement of protons from stroma to thylakoid lumen

(3) Towards the stroma side of thylakoid membrane

(4) Towards the cytoplasm through outer membrane

108. Choose the **correct** statement.

(1) Thylakoid membranes lack NADP reductase enzyme and PS-I

(2) Membrane or lamellae of the grana have both PS II and PS I

(3) PS I and PS II are not interconnected through electron carriers

(4) NADP reductase enzyme is located on lumen side of thylakoid membrane

109. During splitting of water the protons accumulate

(1) In the intermembrane space of chloroplasts

(2) Outside the chloroplast i.e. in the cytoplasm of cell

(3) In the lumen of the thylakoid

(4) In stroma of the chloroplast

110. Which of the following steps of the C_3-cycle is crucial step to continue the cycle uninterrupted?

(1) Reduction

(2) Carboxylation

(3) Regeneration

(4) Decarboxylation

111. Which of the following processes does **not** contribute in the formation of proton gradient across the thylakoid membrane?

(1) Photolysis of water towards lumen

(2) Transfer of H^+ from stroma to lumen

(3) Movement of H^+ from lumen to stroma through CF_0 of ATP synthase

(4) Reduction of $NADP^+$ towards the stroma

112. Cyclic photophosphorylation does **not**
(1) Result in synthesis of ATP
(2) Result in synthesis of $NADH + H^+$
(3) Involve PSI
(4) Occur when light of wavelength beyond 680 nm is available for excitation

113. How many ATP molecule(s) is/are required in regeneration step to form 1 RuBP in Calvin cycle?
(1) Two
(2) Five
(3) One
(4) Six

114. The final electron acceptor in non-cyclic photophosphorylation is
(1) H_2O
(2) ADP
(3) $NADP^+$
(4) NAD^+

115. Chlorophyll-a molecule
(1) Is also known as shield pigment
(2) Shows absorption maxima in blue and red regions of solar radiation
(3) Is yellow-green coloured in the chromatogram
(4) Is present in higher plants only

116. If we take one molecule of α-ketoglutaric acid as respiratory substrate then, how many molecules of ATP will be formed after its complete oxidation?
(1) 1
(2) 3
(3) 9
(4) 8

117. Select the **incorrect** statement w.r.t TCA cycle in eukaryotes.
(1) The cycle starts with the condensation of acetyl group with OAA and water
(2) TCA cycle is confined to mitochondrial matrix
(3) This cycle serves as a pathway for oxidation of carbohydrates only
(4) Succinate dehydrogenase involved in this process is found attached to inner mitochondrial membrane

118. In which of the following reactions of Krebs' cycle, substrate level phosphorylation takes place?
a. Malic acid → Oxaloacetic acid
b. PEP → Pyruvate
c. Succinyl CoA → Succinic acid
d. 3PGA → 2PGA
(1) a & c only
(2) b & d only
(3) b, c & d
(4) c only

119. Proton gradient required for phosphorylation in respiration is created by
(1) The energy of oxidation-reduction
(2) Uphill movement of electrons
(3) The energy of ATP
(4) Splitting of water molecule

120. Complex IV of mitochondria involved in ETS is associated with all, **except**
(1) Cytochrome c oxidase
(2) Cytochrome a & a_3
(3) Cytochrome bc_1
(4) Two copper centres

121. Select the **wrong** statement regarding crucial events in aerobic respiration
(1) Passing on of the electrons removed as part of the hydrogen atoms to molecular O_2 through a channel of other molecules
(2) Complete oxidation of pyruvate
(3) Removal of phosphate from triose phosphate
(4) Stepwise removal of all the hydrogen atoms from pyruvate

122. Each molecule of acetyl CoA produces how many $NADH + H^+$ and $FADH_2$ in the citric acid cycle?
(1) 4 $NADH + H^+$ and 1 $FADH_2$
(2) 3 $NADH + H^+$ and 1 $FADH_2$
(3) 8 $NADH + H^+$ and 2 $FADH_2$
(4) 5 $NADH + H^+$ and 1 $FADH_2$

123. Maximum energy is released during conversion of
(1) Glucose into pyruvic acid
(2) Glucose into ethyl alcohol and CO_2
(3) Pyruvic acid into acetyl CoA
(4) Pyruvic acid into CO_2 and H_2O

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4. Which of the given serves as common oxidative metabolic pathway for carbohydrate, fats and proteins?
(1) Glycolysis
(2) Oxidative decarboxylation
(3) Tricarboxylic acid cycle
(4) Photo-phosphorylation

125. Match the following column-I with column-II.

	Column-I		Column-II
a.	Auxin	(i)	Terpenes
b.	ABA	(ii)	Gas
c.	GA_3	(iii)	Carotenoids
d.	Ethylene	(iv)	Indole compound

Select the **correct** option
(1) a(ii), b(i), c(iii), d(iv)
(2) a(iii), b(iv), c(ii), d(i)
(3) a(iv), b(iii), c(i), d(ii)
(4) a(i), b(iii), c(iv), d(ii)

126. Choose the **incorrect** statement w.r.t. ethylene
(1) Promotes horizontal growth of seedlings
(2) One of the most widely used hormone in agriculture
(3) Promotes germination in peanut seeds but inhibits sprouting of potato tubers
(4) Promotes senescence and abscission in plant organs

127. Efficiency index is the measure of
(1) Ability of the plant to produce new plant material
(2) Time of growth
(3) Number of cells that lose capacity to divide
(4) Rate of differentiation of plant cells

128. All of the following options represent the reasons which cause seed dormancy, **except**
(1) Impermeable and hard seed coat
(2) Presence of Gibberellic acid
(3) Presence of phenolic acids
(4) Immature embryos

129. Heterophyllous development due to environment is found in
(1) Cotton
(2) Buttercup
(3) Coriander
(4) Larkspur

130. Measurement and comparison of total growth per unit time is called
(1) Relative growth rate
(2) Absolute growth rate
(3) Arithmetic growth rate
(4) Geometric growth rate

131. The process through which ATP is synthesized by cells (in mitochondria and chloroplasts) is named
(1) Carboxylation
(2) Decarboxylation
(3) Reduction
(4) Phosphorylation

132. All of the following scientists gave the scheme of glycolysis, **except**
(1) Gustav Embden
(2) Hans Kreb
(3) Otto Meyerhof
(4) J. Parnas

133. Invertase catalyses the conversion of
(1) Glucose → Glucose-6-phosphate
(2) Fructose-6-phosphate → Fructose 1, 6-bisphosphate
(3) Glyceraldehyde-3-phosphate → Dihydroxyacetone phosphate
(4) Sucrose → Glucose + Fructose

134. During the conversion of 3-phosphoglyceraldehyde to 1, 3-bisphosphoglycerate
(1) ATP is consumed
(2) ATP is produced
(3) Two redox-equivalents are removed
(4) CO_2 is released

135. The key product of glycolysis is
(1) Glucose
(2) Fructose
(3) PEP
(4) Pyruvic acid

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136. Which of the following processes does **not** yield CO_2?
(1) Oxidative decarboxylation of pyruvate
(2) Krebs cycle
(3) Lactic acid fermentation
(4) Alcohol fermentation

137. Pyruvic acid decarboxylase is an enzyme involved in the process of
(1) Glycolysis
(2) Lactic acid fermentation
(3) Alcohol fermentation
(4) Tricarboxylic acid cycle

138. All of the following processes yield NAD^+, **except**
(1) Lactic acid fermentation
(2) Alcohol fermentation
(3) Citric acid cycle
(4) ETS and oxidative phosphorylation

139. The most favoured substrate for respiration is
(1) Amino acid
(2) Lipids
(3) Glucose
(4) DNA

140. How many molecules of CO_2 are released when two molecules of tripalmitin are used as a respiratory substrate?
(1) 98
(2) 102
(3) 145
(4) 260

141. Read the following statements and choose the **correct** option.
Statement A: Vernalisation prevents precocious reproductive development late in the growing season.
Statement B: Subjecting the growing of a biennial plant to a cold treatment stimulates a subsequent photoperiodic flowering response.
(1) Only statement A is correct
(2) Only statement B is correct
(3) Both the statements A and B are correct
(4) Both the statements A and B are incorrect

SECTION-B

142. Which of the following statements is **incorrect** regarding photoperiodism in plants?
(1) The hormonal substance responsible for flowering migrates from shoot apices to leaves to induce flowering
(2) Shoot apices of plants cannot perceive photoperiods
(3) Duration of dark period is also important for flowering
(4) The critical duration of light and dark is different for different plants

143. Bundle sheath cells in C_4 plants are characterised by how many of the following features given below?
i. Large cells
ii. Having large number of chloroplasts
iii. Thin walled
iv. Impervious to gaseous exchange
v. Presence of intercellular spaces
(1) (i), (ii) and (iii) only
(2) (ii) and (iv) only
(3) (i), (iii) and (v) only
(4) (i), (ii) and (iv) only

144. Law of limiting factors was given by
(1) Cornelius van Niel
(2) Julius von Sachs
(3) Joseph Priestley
(4) Blackman

145. Chemiosmosis does **not** require
(1) Electron gradient
(2) Membrane
(3) Proton pump
(4) Proton gradient

146. The location in which dark reactions of photosynthesis take place in chloroplast is
(1) Lumen of thylakoid
(2) Intermembrane space
(3) Stroma lamellae
(4) Stroma

147. Which of the following pair of organisms were used by T.W. Engelmann for his experiment?
(1) Mouse and mint plant
(2) Aquatic plant and mint plant
(3) Purple sulphur bacteria and cyanobacteria
(4) *Cladophora* and aerobic bacteria

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8. Which of the following equations correctly depicts arithmetic growth?

(1) $W_1 = W_0 e^{rt}$

(2) $W_1 = W_0 rt$

(3) $L_t = L_0 + rt$

(4) $L_t = \frac{L_0}{rt}$

149. Closure of stomata is stimulated by

(1) Gibberellin

(2) Abscisic acid

(3) Ethylene

(4) Auxin

150. Which of the following option is **not** a feature of cells in meristematic zone?

(1) Primary cellulosic cell wall

(2) Abundant plasmodesmatal connections

(3) Cells are rich in protoplasm

(4) Increased vacuolation

ZOOLOGY

SECTION-A

151. *Crocodilus* and *Ophiosaurus* are similar to each other in the following features, **except**

(1) Four-chambered heart

(2) Sexes are separate

(3) Oviparous

(4) Pneumatic bones

152. Match column I with column II and select the **correct** option.

	Column I		Column II
(a)	Metamerism	(i)	*Sycon*
(b)	Cnidoblasts	(ii)	*Pleurobrachia*
(c)	Ciliated comb plates	(iii)	*Nereis*
(d)	Water canal system	(iv)	*Meandrina*

(1) a(i), b(ii), c(iii), d(iv)

(2) a(iii), b(iv), c(ii), d(i)

(3) a(ii), b(iii), c(iv), d(i)

(4) a(iv), b(iii), c(ii), d(i)

153. Consider the following features w.r.t cyclostomes.

(a) They have sucking and circular mouth without jaws.

(b) Elongated body bearing 6-15 pairs of gill slits for respiration

(c) The body is devoid of scales and unpaired fins

(d) Circulation is of open type

Choose the **correct** option.

(1) (a) and (b) are correct

(2) (c) and (d) are correct

(3) (a), (b) and (c) are incorrect

(4) (d) is correct

154. Which of the following is an incorrectly matched pair of a body feature and the animal possessing it?

(1) Operculum – *Betta*

(2) Tympanum – *Ichthyophis*

(3) Pneumatic bones – *Chameleon*

(4) Radula – *Dentalium*

155. Which of the following features is not present in the 'King crab"?

(1) Jointed appendages

(2) Bilateral symmetry

(3) Organ system level of organisation

(4) Parapodia

156. How many of the following organisms show scattered pouches of mesoderm in between the ectoderm and endoderm in the body cavity?

Tapeworm, Roundworm, Filarial worm, Hookworm

(1) One
(2) Two
(3) Three
(4) Four

157. Which of the following is the common characteristics of all vertebrates without exception?
(1) Body is divisible into head, trunk and tail
(2) Two pairs of functional appendages
(3) Presence of notochord
(4) Direct development

158. Identify the group of animals characterized by crop and gizzard in their digestive system. Choose the correct option.
(1) *Aptenodytes, Pteropus, Neophron*
(2) *Pavo, Psittacula, Periplaneta*
(3) *Corvus, Columba, Crocodilus*
(4) *Bufo, Delphinus, Struthio*

159. Which one of the following pair of animals comprises 'Jawed fishes'?
(1) Hagfish and Lamprey
(2) Magur and Sting ray
(3) Hagfish and Devil fish
(4) Squid and Flying fish

160. Choose the option which correctly matches its example, two characteristics and phylum.

	Example	Two characters	Phylum
(1)	*Ophiura*	(a) Water canal system (b) Spiny bodied	Echinodermata
(2)	*Laccifer*	(a) Jointed appendages (b) Chitinous endoskeleton	Arthropoda
(3)	*Ancylostoma*	(a) Muscular pharynx (b) Dioecious	Aschelminthes
(4)	*Fasciola*	(a) Hooks and suckers present (b) Flame cells	Platyhelminthes

(1) (1)
(2) (2)
(3) (3)
(4) (4)

161. A pair of elongated, pink coloured sac-like structure present in the upper part of trunk region in *Rana tigrina* is
(1) Lungs
(2) Tympanum
(3) Nostrils
(4) Oesophagus

162. Which of the following is mismatched w.r.t. adult *Rana tigrina*?
(1) Length of alimentary canal : Short
(2) Fertilisation : External
(3) Organ of balancing and hearing : Ear
(4) Vasa efferentia : 10-12 pairs in each testis

163. How many of the following are associated with circulatory system of frog?
(a) Heart is a muscular structure situated in upper part of the body cavity.
(b) A circular structure called sinus venosus joins the right atrium.
(c) The ventricle open into a sac like conus arteriosus on the dorsal side of the heart.
(d) The blood from the heart is carried to all parts of the body by the veins.
Select the **correct** option.
(1) One
(2) Two
(3) Three
(4) Four

164. Which of the following is not **incorrect** w.r.t. excretory system of a male frog?
(1) Kidneys are compact, dark red and bean like structures situated a little anteriorly in the body cavity on both side of vertebral column
(2) Kidneys are composed of several structural and functional units called uriniferous tubules.
(3) The ureters enter into kidney and act as urinogenital duct.
(4) The ureters and genital duct open separately in the cloaca

165. Frog erythrocytes are
(1) Nucleated and without haemoglobin
(2) Enucleated and with haemoglobin
(3) Nucleated and with haemoglobin
(4) Enucleated and without haemoglobin

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6. Which of the following set of sense organs are not well -organised structures but are cellular aggregations around nerve endings in the frogs?

(1) Sensory papillae, taste buds, internal ears

(2) Eyes, nasal epithelium, internal ears

(3) Sensory papillae, taste buds, nasal epithelium

(4) Simple eyes, internal ears, taste buds

167. Cranial nerve arising from the brain in female frogs are

(1) 20 pairs

(2) 10 pairs

(3) 2 pairs

(4) 5 pairs

168. In *Rana tigrina*, the hindlimbs end in 'B' digits and forelimbs end in 'A' digits. Choose the option that fills the blanks **correctly**.

	A	B
(1)	Five	Five
(2)	Four	Five
(3)	Five	Four
(4)	Four	Four

(1) (1)

(2) (2)

(3) (3)

(4) (4)

169. Complete the analogy and select the **correct** option w.r.t. adult frog.

Aestivation : Cutaneous respiration : : Hibernation : _____

(1) Pulmonary respiration

(2) Branchial respiration

(3) Cutaneous respiration

(4) Buccopharyngeal respiration

170. How many statements given below are not related to female frog?

(a) Reproductive organs include a pair of ovaries.

(b) Absence of copulatory pad on the first digit of the fore limbs.

(c) Presence of sound producing vocal sacs.

(d) The ovaries are situated near kidneys and there is functional connection with kidneys.

(e) Mature female can lay 2500-3000 ova at a time

Select the **correct** option.

(1) Three

(2) Two

(3) One

(4) Four

171. Match column I and column II and select the **correct** option.

	Column I		Column II
(a)	Physical barriers	(i)	Interferons
(b)	Physiological barriers	(ii)	Tears from eyes
(c)	Cellular barriers	(iii)	Skin
(d)	Cytokine barriers	(iv)	NK cells

(1) a(iii), b(ii), c(iv), d(i)

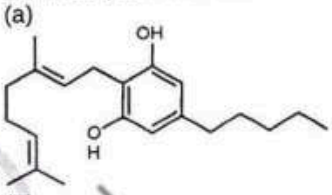
(2) a(i), b(ii), c(iii), d(iv)

(3) a(ii), b(iii), c(i), d(iv)

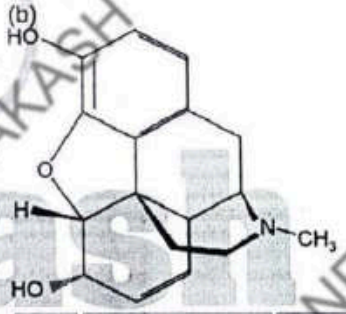
(4) a(iii), b(ii), c(i), d(iv)

172. Identify the molecule (a) and (b) shown in the below given figures and select the **correct** option w.r.t. them.

(a)



(b)



	Drug	Receptors	Source
(1)	(a) Cannabinoids	CNS	Coca plant
(2)	(a) Cannabinoids	Principally present in brain	Inflorescence, flower tops, leaves and resin of *Cannabis*
(3)	(b) Opioids	CNS, GIT	Latex of *Atropa*
(4)	(b) Opioids	CNS, GIT	*Erythroxylum coca*

(1) (1)

(2) (2)

(3) (3)

(4) (4)

173. How many disulphide bonds are present in structure of an antibody molecule represented as H_2L_2?

(1) 16

(2) 12

(3) 32

(4) 10

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174. Which one of the following options gives the correct matching of a disease with its causative organism and mode of infection?

	Disease	Causative organisms	Mode of infection
(1)	Ringworm	*Microsporum*	Contaminated food and water
(2)	Pneumonia	*Haemophilus influenzae*	Droplet infection
(3)	Typhoid	*Salmonella typhi*	Direct contact with skin
(4)	Common cold	Rhino virus	Vector borne

(1) (1)
(2) (2)
(3) (3)
(4) (4)

175. **Assertion (A):** Macrophages act like a HIV factory.
Reason (R): The HIV virus enters into macrophages where RNA genome of the virus replicates to form viral RNA with the help of the enzyme reverse transcriptase.
In the light of above statement, choose the correct option.
(1) Both (A) and (R) are correct and (R) is correct explanation of (A).
(2) Both (A) and (R) are false.
(3) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(4) (A) is true and (R) is false.

176. Complete the analogy and choose the **correct** option.
X-rays : Ionising radiations : : Gamma rays : _____
(1) Non-ionising radiations
(2) Cellular oncogenes
(3) Ionising radiations
(4) Proto-oncogenes

177. Treatment of AIDS is only partially effective with
(1) Anti-retroviral drugs
(2) Antibiotics
(3) Gene therapy
(4) Surgery

178. A female athlete started using anabolic steroids to enhance her performance. After a few months of consumption she noticed some changes in her body. How many of the below given changes could be observed due to intake of anabolic steroids.?
(a) Masculinisation
(b) Increased aggressiveness
(c) Enlargement of clitoris
(d) Reduction in size of spermatozoa
(e) Breast enlargement
(f) Enlargement of prostate gland
Select the **correct** option.
(1) Six
(2) Three
(3) Two
(4) Four

179. Which of the following is not **incorrect** abbreviations?
(1) NACO – National AIDS and Cancer Organisation
(2) MALT – Mucosa Associated Lymphoid Tissue
(3) ELISA – Enzyme Linked Immuno Sorbent Association
(4) AIDS – Acquired Immuno-Deficiency System

180. Choose the **odd** one from the following.
(1) Auto-immune disease – Allergy
(2) Humoral immune response – Graft rejection
(3) Primary immune response – Anamnestic response
(4) Colostrum – Passive immunity to new born

181. Which one of the following animals is not oviparous?
(1) *Chaetopleura*
(2) *Hippocampus*
(3) *Carcharodon*
(4) *Calotes*

182. Proboscis gland is an excretory structure found in
(1) *Locusta*
(2) *Asterias*
(3) *Balanoglossus*
(4) *Chaetopleura*

183. Select the option in which animals belong to the same phylum.
(1) Apple snail, devil fish, king crab
(2) Cuttle fish, starfish, sea lily
(3) *Ascidia*, *Salpa*, *Doliolum*
(4) Lancelet, *Myxine*, *Balanoglossus*

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184. Closed type of circulatory system is present in members of all the following taxa, **except**
(1) Annelida
(2) Arthropoda
(3) Vertebrata
(4) Chordata

185. Read the following statements A and B.
Statement A : Chondrichthyes show internal fertilization and many of them are viviparous.
Statement B : Due to the absence of air bladder, chondrichthyes have to swim constantly to avoid sinking.
Choose the **correct** option.
(1) Statement A is correct but B is incorrect.
(2) Statement A is incorrect but B is correct.
(3) Both statements are correct.
(4) Both statements are incorrect.

SECTION-B

186. A marine fish that possesses terminal mouth and lacks the capacity to regulate their body temperature is
(1) *Scoliodon*
(2) *Exocoetus*
(3) *Pristis*
(4) *Myxine*

187. Choose the **odd** one w.r.t. symmetry.
(1) *Adult Asterias*
(2) *Pleurobrachia*
(3) *Hydra*
(4) *Planaria*

188. Choose the **correct** match w.r.t. general reptilian characters.
(1) Fertilisation - External
(2) Development - Indirect
(3) Skin - Dry and cornified
(4) Heart - Two chambered

189. Which amongst the following **cannot** be a point of difference between chondrichthyes and osteichthyes?
(1) Location or position of mouth
(2) Types of scales
(3) Presence of operculum
(4) Cold-blooded nature

190. Select the **correct route** for the passage of sperms in male frogs
(1) Testes → Bidder's canal→ Kidney → vasa efferentia → urinogenital duct → cloaca
(2) Testes → vasa efferentia → Kidney → seminal vesicle→ urinogenital duct → cloaca
(3) Testes → Bidder's canal → vasa efferentia → cloaca→ ureter
(4) Testes → vasa efferentia → Kidney → Bidder's canal → urinogenital duct → cloaca

191. Match the following structures in column I and their location in column II w.r.t. *Pheretima*.

	Column I		Column II
a.	Female genital pore	(i)	10th and 11th segments
b.	Male genital pores	(ii)	12th and 13th segments
c.	Ovaries	(iii)	18th segment
d.	Testes	(iv)	14th segment

Choose the **correct** option.
(1) a(iv), b(iii), c(i), d(ii)
(2) a(iv), b(iii), c(ii), d(i)
(3) a(iv), b(ii), c(iii), d(i)
(4) a(iv), b(ii), c(i), d(iii)

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192. Who disproved the "Good Humor" hypothesis of health?
(1) Hippocrates
(2) William Harvey
(3) Stanley Cohen
(4) Herbert Boyer

193. Cell-mediated immune response is mediated by
(1) B-lymphocytes
(2) T-lymphocytes
(3) Monocytes
(4) Macrophages

194. All of the following options are associated with detection of cancer, **except**
(1) Biopsy
(2) Computed tomography
(3) Radiotherapy
(4) MRI

195. Withdrawal symptom is seen in the body when
(1) Dose of drugs / alcohol is abruptly increased
(2) A person is under unduly peer pressure
(3) There is sudden discontinuation of drugs / alcohol
(4) Allergic response is generated upon intake of alcohol / tobacco

196. Sporozoites are infective stage of *Plasmodium* for humans and are produced in infected
(1) Human liver
(2) Human RBC
(3) Female *Anopheles* mosquito
(4) Male *Anopheles* mosquito

197. In _____, notochord is present only in larval tail, while in _____, it extends from head to tail region.
Select the **correct** option that fills the blanks.
(1) Urochordates, Cephalochordates
(2) Hemichordates, Urochordates
(3) Tunicates, Hemichordates
(4) Cephalochordates, Cyclostomates

198. _____ are transparent, membranous and are used for flight in cockroaches.
Fill the blank with the **correct** option
(1) Tegmina
(2) Elytra
(3) Hindwings
(4) Forewings

199. The development of *P. americana* is
(1) Ametabolous
(2) Hypermetabolous
(3) Holometabolous
(4) Paurometabolous

200. In frog, chyme formation occurs in
(1) Intestine
(2) Stomach
(3) Liver
(4) Rectum

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